A Different Perspective on Derivatives

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1. Introduction

In this paper, we look at a different way of calculating derivatives, one that departs from the First Principles method. We will mostly deal with simple derivatives, and the aim is simply to offer a fresh point of view.

Keep in mind

This method is **less rigorous** than the First Principles approach. It is meant to build intuition, not to replace formal proofs.

2. The Perspective

We will learn this perspective through examples, starting with a simple one.

Example: $y = x^2$

Suppose we want to differentiate $y = x^2$, that is, find $\dfrac{dy}{dx}$.

Let x change by a tiny amount dx, and let y change correspondingly by dy. Then

$$y + dy = (x + dx)^2$$
$$y + dy = x^2 + 2x\,dx + (dx)^2.$$

Since $y = x^2$, subtracting it from both sides gives

$$dy = 2x\,dx + (dx)^2.$$

Now comes the key idea (call it a **concept** or a **workaround**, whichever you prefer). The quantity dx is already an infinitesimally small change in x. Squaring it gives something far smaller still, so $(dx)^2$ is negligible compared with dx. We therefore drop it:

$$dy = 2x\,dx.$$

Dividing both sides by dx,

$$\frac{dy}{dx} = 2x.$$

A note on dividing by dx

Derivatives are not exactly fractions, but they *behave* like fractions in many ways, so dividing by dx is valid here.

We have arrived at our result:

Result

$$\frac{d}{dx}\left(x^2\right) = 2x$$

2.1 A Little Deep Dive

One generalised idea is worth remembering. Higher powers of the differential dx are negligible compared with dx itself (and hence also compared with x). For example,

$$(dx)^3 \approx 0 \quad \text{compared with } dx.$$

Rule of thumb

Whenever a term contains $(dx)^2$, $(dx)^3$, or any higher power of dx, neglect it.

3. To the Reader

Keeping this in mind, try to derive the following results yourself.

Exercises

1. If $y = x^3$, find $\dfrac{dy}{dx}$.
2. If $y = \dfrac{1}{x^2}$, find $\dfrac{dy}{dx}$.
3. Finally, generalise your results to find $\dfrac{d}{dx}\left(x^n\right)$.